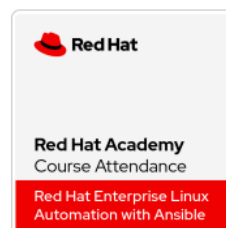




Red Hat Enterprise Linux Automation with Ansible (RH294 - RHA) - Ver. 9.0

ISSUED TO

SANJAY DEVANG



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Verify: <https://www.credly.com/go/6hWHiDOe>

```
student@workstation:~/plaX +
state: started
- name: Test php page is installed
  ansible.builtin.copy:
    src: index.php
    dest: /var/www/html/index.php
    mode: 0644
Sanjay Devang 2025-04-25 > vi internet.yml
Sanjay Devang 2025-04-25 > ansible-navigator run -m stdout internet.yml --syntax-check
[WARNING]: Could not match supplied host pattern, ignoring: workstation
playbook: /home/student/playbook-review/internet.yml
Sanjay Devang 2025-04-25 > ansible-navigator run -m stdout internet.yml

PLAY [Enable internet services] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]

TASK [Latest version of all required packages installed] *****
changed: [serverb.lab.example.com]

TASK [firewalld enabled and running] *****
ok: [serverb.lab.example.com]

TASK [firewalld permits http service] *****
changed: [serverb.lab.example.com]

TASK [httpd enabled and running] *****
changed: [serverb.lab.example.com]

TASK [mariadb enabled and running] *****
changed: [serverb.lab.example.com]

TASK [Test php page is installed] *****
changed: [serverb.lab.example.com]
[WARNING]: Could not match supplied host pattern, ignoring: workstation

PLAY [Test internet web server] *****
skipping: no hosts matched

PLAY RECAP *****
serverb.lab.example.com : ok=7  changed=5  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
Sanjay Devang 2025-04-25 > □
```

Figure 1 Running internet.yml Ansible playbook to configure and enable web services

Note: The playbook `internet.yml` was executed successfully using `ansible-navigator`. It enabled firewall, HTTP, and MariaDB services on `serverb.lab.example.com`. A warning appeared about an unmatched host pattern for `workstation`, but the tasks for `serverb` completed without any errors. The final play recap shows 5 changes and no failures.

SANJAY DEVANG

```
PLAY [Test internet web server] *****
skipping: no hosts matched

PLAY RECAP *****
serverb.lab.example.com : ok=7  changed=5  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
Sanjay Devang 2025-04-25 > lab grade playbook-review

Grading lab.

  • Checking lab systems ..... SUCCESS
  • Ensuring httpd package is installed on serverb ..... PASS
  • Ensuring mariadb-server package is installed on serverb ..... PASS
  • Ensuring php package is installed on serverb ..... PASS
  • Ensuring php-mysqlnd package is installed on serverb ..... PASS
  • Ensuring httpd service is started on serverb ..... PASS
  • Ensuring mariadb service is started on serverb ..... PASS
  • Ensuring http service is enabled in firewalld on serverb ..... PASS
  • Ensuring the web server is reachable on serverb ..... PASS

Overall lab grade: PASS

Sanjay Devang 2025-04-25 > □
```

Figure 2 Lab grading results showing all checks passed successfully.

Note:The lab grading script verified the setup of the web server and related services on `serverb.lab.example.com`. All checks for package installation, service status, firewall settings, and web server reachability passed.

```
Sanjay Devang 2025-04-25 > mkdir -p ~/data-review
Sanjay Devang 2025-04-25 > cd ~/data-review
Sanjay Devang 2025-04-25 > vi ~/data-review/diagnose.yml
Sanjay Devang 2025-04-25 > ansible-navigator run -m stdout --pae false diagnose.yml

PLAY [Diagnose httpd issues] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]

TASK [Check if httpd is installed] *****
ok: [serverb.lab.example.com]

TASK [Check httpd configuration] *****
ok: [serverb.lab.example.com]

TASK [Check if SSL certificates exist] *****
ok: [serverb.lab.example.com] => (item=/etc/pki/tls/certs/localhost.crt)
ok: [serverb.lab.example.com] => (item=/etc/pki/tls/private/localhost.key)

TASK [Show diagnostics] *****
ok: [serverb.lab.example.com] => {
  "msg": [
    "HTTPD package status: False",
    "HTTPD config check: []",
    "HTTPD config errors: ['AH00526: Syntax error on line 5 of /etc/httpd/conf.d/ssl.conf:', 'Cannot define multiple Listeners on the same IP:port']",
    "SSL certificate status: True",
    "SSL key status: True"
  ]
}

PLAY RECAP *****
serverb.lab.example.com : ok=5  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

Sanjay Devang 2025-04-25 > vi ~/data-review/files/httpd.conf
Sanjay Devang 2025-04-25 > vi ~/data-review/fix-httpd.yml
Sanjay Devang 2025-04-25 > ansible-navigator run -m stdout --pae false playbook.yml --vault-id @prompt
Vault password (default):

PLAY [Install and configure webserver with basic auth] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]
```

Figure 3 Diagnosing HTTPD service and SSL configuration issues using `diagnose.yml`.

Note:The `diagnose.yml` playbook was executed to check the HTTPD package, config, and SSL certificates. A syntax error in `ssl.conf` was detected, specifically mentioning multiple listeners on the same IP:port. The SSL certificates were confirmed as valid. After diagnosis, preparations were made to fix the issue using a secure `playbook.yml` with Ansible Vault authentication.


```
Sanjay Devang 2025-04-25 > lab start control-review
```

```
Starting lab.
```

- Checking lab systems SUCCESS
- Adding exercise content SUCCESS
- Ensuring required packages are installed SUCCESS

```
Sanjay Devang 2025-04-25 > cd ~/control-review
```

```
Sanjay Devang 2025-04-25 > LS
```

```
bash: LS: command not found...
```

```
Similar command is: 'ls'
```

```
Sanjay Devang 2025-04-25 > cd ~/control-review
```

```
Sanjay Devang 2025-04-25 > LS
```

```
bash: LS: command not found...
```

```
Similar command is: 'ls'
```

```
Sanjay Devang 2025-04-25 > ls
```

```
ansible.cfg index.html inventory playbook.yml server.crt server.key ssl.conf vars.yml
```

```
Sanjay Devang 2025-04-25 > vi playbook.yml
```

```
Sanjay Devang 2025-04-25 > ansible-navigator run -m stdout playbook.yml
```

```
PLAY [Playbook Control Lab] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [serverb.lab.example.com]
```

```
TASK [Show failed system requirements message] *****
```

```
skipping: [serverb.lab.example.com]
```

```
TASK [Ensure required packages are present] *****
```

```
changed: [serverb.lab.example.com]
```

```
TASK [Ensure services are started and enabled] *****
```

```
changed: [serverb.lab.example.com] => (item=httpd)
```

```
ok: [serverb.lab.example.com] => (item=firewalld)
```

```
TASK [Create SSL cert directory] *****
```

```
changed: [serverb.lab.example.com]
```

```
TASK [Copy config files] *****
```

```
changed: [serverb.lab.example.com] => (item={'src': 'server.key', 'dest': '/etc/httpd/conf.d/ssl'})
```

```
changed: [serverb.lab.example.com] => (item={'src': 'server.crt', 'dest': '/etc/httpd/conf.d/ssl'})
```

```
changed: [serverb.lab.example.com] => (item={'src': 'ssl.conf', 'dest': '/etc/httpd/conf.d'})
```

```
changed: [serverb.lab.example.com] => (item={'src': 'index.html', 'dest': '/var/www/html'})
```

Figure 4 Executing `playbook.yml` for the Control Review Lab setup.

Note:The Control Review Lab was initiated and the `playbook.yml` was run to ensure services and configurations were correctly applied on `serverb.lab.example.com`. Key tasks included installing necessary packages, starting services like HTTPD and firewall, creating the SSL directory, and copying configuration files. Minor shell command typos were encountered but corrected without impact.

```

student@workstation:~/cor X +
* issuer: C=US; L=Default City; O=Red Hat; OU=Training; CN=serverb.lab.example.com
* SSL certificate verify result: self-signed certificate (18), continuing anyway.
* TLSv1.2 (OUT), TLS header, Unknown (23):
> GET / HTTP/1.1
> Host: serverb.lab.example.com
> User-Agent: curl/7.76.1
> Accept: */*
>
* TLSv1.2 (IN), TLS header, Unknown (23):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.2 (IN), TLS header, Unknown (23):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* old SSL session ID is stale, removing
* TLSv1.2 (IN), TLS header, Unknown (23):
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< Date: Fri, 25 Apr 2025 18:08:37 GMT
< Server: Apache/2.4.51 (Red Hat Enterprise Linux) OpenSSL/3.0.1
< Last-Modified: Fri, 25 Apr 2025 18:08:28 GMT
< ETag: "24-6339e38a95e88"
< Accept-Ranges: bytes
< Content-Length: 36
< Content-Type: text/html; charset=UTF-8
<
Configured for both HTTP and HTTPS.
* Connection #0 to host serverb.lab.example.com left intact
Sanjay Devang 2025-04-25 > lab grade control-review

Grading lab.

  • Checking lab systems ..... SUCCESS
  • Ensuring Ansible playbook is present ..... PASS
  • Ensuring httpd package is installed on serverb ..... PASS
  • Ensuring mod_ssl package is installed on serverb ..... PASS
  • Ensuring httpd service is started on serverb ..... PASS
  • Ensuring firewall service is started on serverb ..... PASS
  • Ensuring SSL configuration file is present ..... PASS
  • Ensuring SSL certificate file is present ..... PASS
  • Ensuring SSL key file is present ..... PASS
  • Ensuring the web server is reachable on serverb ..... PASS

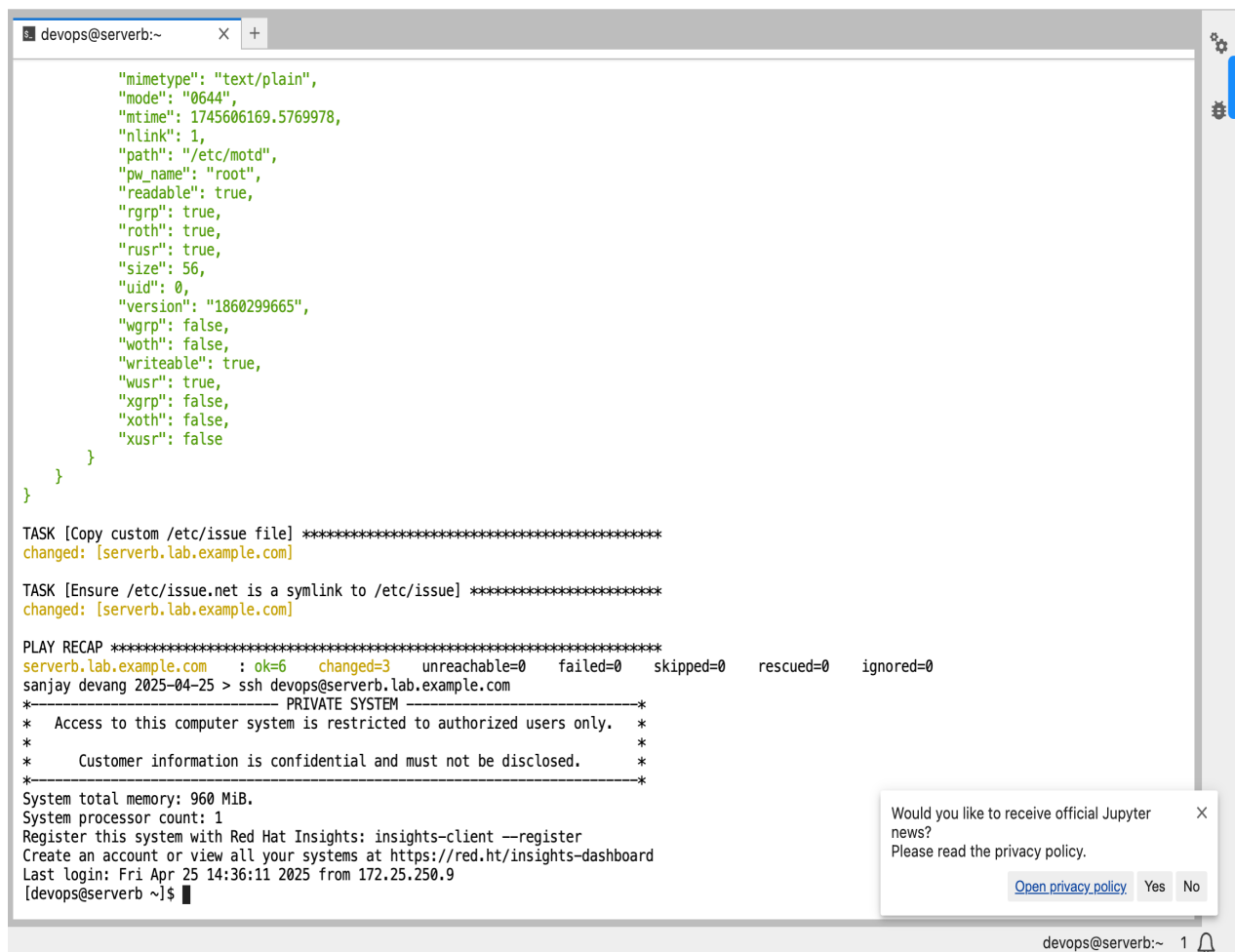
Overall lab grade: PASS

Sanjay Devang 2025-04-25 >

```

Figure 5 SSL connection verification and successful grading of the Control Review Lab.

Note:An HTTPS connection was established to `serverb.lab.example.com`, confirming proper SSL configuration despite a self-signed certificate warning. The Control Review Lab grading validated all key requirements including HTTPD setup, SSL files, firewall settings, and server reachability.



```
devops@serverb:~  
  "mimetype": "text/plain",  
  "mode": "0644",  
  "mtime": 1745606169.5769978,  
  "nlink": 1,  
  "path": "/etc/motd",  
  "pw_name": "root",  
  "readable": true,  
  "rgrp": true,  
  "roth": true,  
  "rusr": true,  
  "size": 56,  
  "uid": 0,  
  "version": "1860299665",  
  "wgrp": false,  
  "woth": false,  
  "writeable": true,  
  "wusr": true,  
  "xgrp": false,  
  "xoth": false,  
  "xusr": false  
}  
}  
}  
  
TASK [Copy custom /etc/issue file] *****  
changed: [serverb.lab.example.com]  
  
TASK [Ensure /etc/issue.net is a symlink to /etc/issue] *****  
changed: [serverb.lab.example.com]  
  
PLAY RECAP *****  
serverb.lab.example.com : ok=6  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0  
sanjay devang 2025-04-25 > ssh devops@serverb.lab.example.com  
----- PRIVATE SYSTEM -----  
* Access to this computer system is restricted to authorized users only. *  
* Customer information is confidential and must not be disclosed. *  
-----  
System total memory: 960 MiB.  
System processor count: 1  
Register this system with Red Hat Insights: insights-client --register  
Create an account or view all your systems at https://red.ht/insights-dashboard  
Last login: Fri Apr 25 14:36:11 2025 from 172.25.250.9  
[devops@serverb ~]$
```

Would you like to receive official Jupyter news?
Please read the privacy policy.

[Open privacy policy](#) Yes No

devops@serverb:~ 1

Figure 6 Custom /etc/issue banner file deployed successfully on the server

Note:A customized `/etc/issue` file was copied and enforced on `serverb.lab.example.com`, displaying a privacy and security warning message at login. The Ansible playbook tasks completed successfully with no errors. The server now shows a secure login banner message to users, enhancing system compliance and security awareness.

```

student@workstation:~/file X +

    "xgrp": false,
    "xoth": false,
    "xusr": false
  }
}
}

TASK [Copy custom /etc/issue file] *****
changed: [serverb.lab.example.com]

TASK [Ensure /etc/issue.net is a symlink to /etc/issue] *****
changed: [serverb.lab.example.com]

PLAY RECAP *****
serverb.lab.example.com : ok=6  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
sanjay devang 2025-04-25 > ssh devops@serverb.lab.example.com
*----- PRIVATE SYSTEM -----*
* Access to this computer system is restricted to authorized users only. *
* * *
* Customer information is confidential and must not be disclosed. *
*-----*
System total memory: 960 MiB.
System processor count: 1
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Fri Apr 25 14:36:11 2025 from 172.25.250.9
[devops@serverb ~]$ lab grade file-review
-bash: lab: command not found
[devops@serverb ~]$ logout
Connection to serverb.lab.example.com closed.
sanjay devang 2025-04-25 > lab grade file-review

Grading lab.

· Checking lab systems ..... SUCCESS
· Ensuring motd.j2 file is present ..... PASS
· Ensuring motd.yml playbook is present ..... PASS
· Checking /etc/motd ..... PASS
· Checking /etc/issue ..... PASS
· Ensuring /etc/issue.net symlink is present ..... PASS

Overall lab grade: PASS

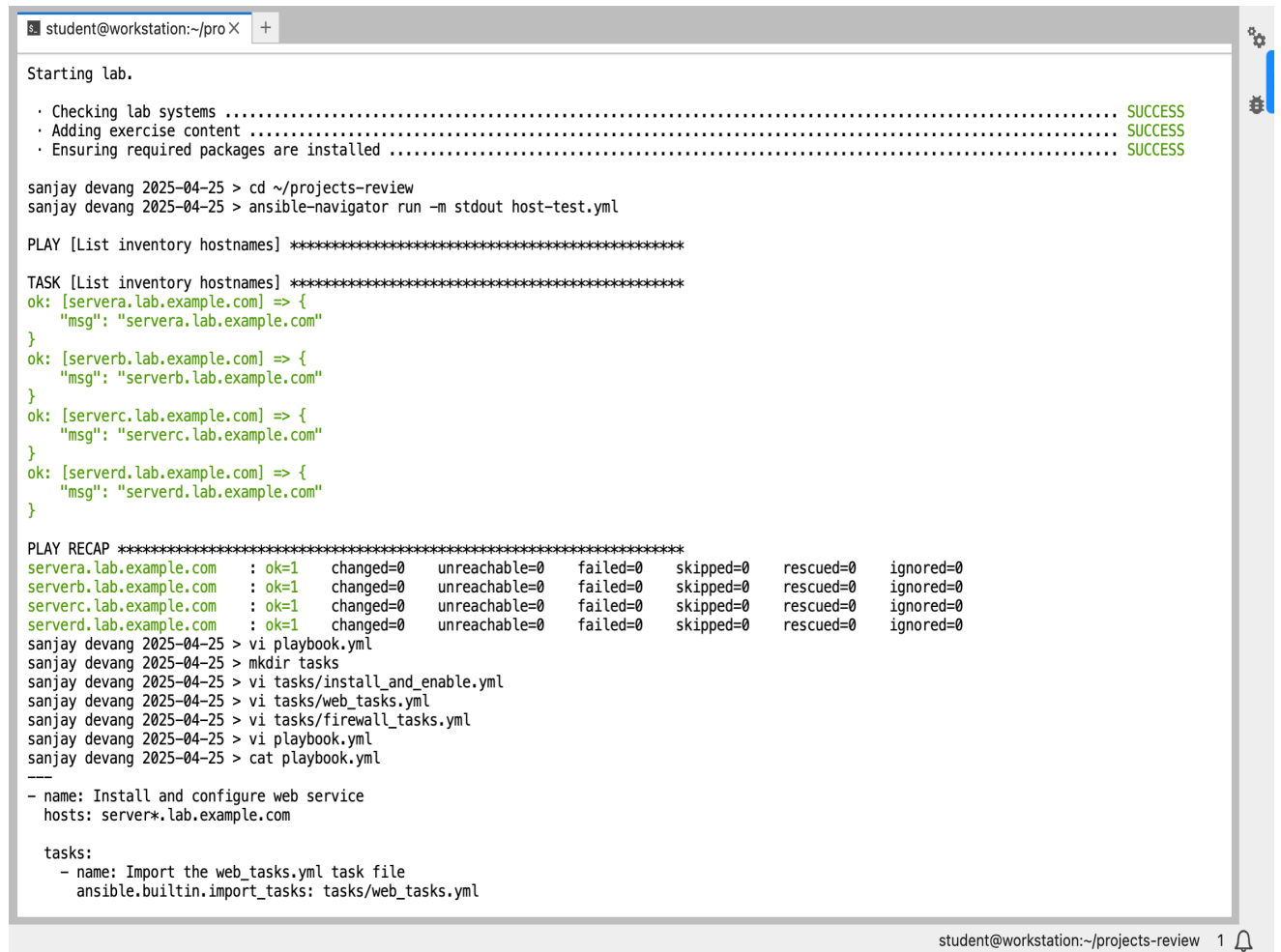
sanjay devang 2025-04-25 >

```

student@workstation:~/file-review 1

Figure 7 Final lab grading for file review task confirming banner file setup success.

Note: The file review lab checked for the presence and correct deployment of the custom `/etc/issue` and `/etc/issue.net` files along with their symlink.



```

student@workstation:~/pro X +
Starting lab.
  · Checking lab systems ..... SUCCESS
  · Adding exercise content ..... SUCCESS
  · Ensuring required packages are installed ..... SUCCESS

sanjay devang 2025-04-25 > cd ~/projects-review
sanjay devang 2025-04-25 > ansible-navigator run -m stdout host-test.yml

PLAY [List inventory hostnames] *****

TASK [List inventory hostnames] *****
ok: [servera.lab.example.com] => {
  "msg": "servera.lab.example.com"
}
ok: [serverb.lab.example.com] => {
  "msg": "serverb.lab.example.com"
}
ok: [serverc.lab.example.com] => {
  "msg": "serverc.lab.example.com"
}
ok: [serverd.lab.example.com] => {
  "msg": "serverd.lab.example.com"
}

PLAY RECAP *****
servera.lab.example.com : ok=1  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverb.lab.example.com : ok=1  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverc.lab.example.com : ok=1  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverd.lab.example.com : ok=1  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

sanjay devang 2025-04-25 > vi playbook.yml
sanjay devang 2025-04-25 > mkdir tasks
sanjay devang 2025-04-25 > vi tasks/install_and_enable.yml
sanjay devang 2025-04-25 > vi tasks/web_tasks.yml
sanjay devang 2025-04-25 > vi tasks/firewall_tasks.yml
sanjay devang 2025-04-25 > vi playbook.yml
sanjay devang 2025-04-25 > cat playbook.yml
---
- name: Install and configure web service
  hosts: server*.lab.example.com

  tasks:
    - name: Import the web_tasks.yml task file
      ansible.builtin.import_tasks: tasks/web_tasks.yml

```

Figure 8 Inventory listing and task file setup for project review lab.

Note:The `host-test.yml` playbook successfully listed all target inventory hosts: `servera`, `serverb`, `serverc`, and `serverd.lab.example.com`. After verifying hosts, multiple task files (`install_and_enable.yml`, `web_tasks.yml`, `firewall_tasks.yml`) were created under a `tasks` directory. The main `playbook.yml` was prepared to import these task files for web service configuration.

```

student@workstation:~/pro X +

ok: [serverb.lab.example.com]
ok: [serverc.lab.example.com]
ok: [servera.lab.example.com]
ok: [serverd.lab.example.com]

TASK [Enable and start firewall] *****
ok: [serverb.lab.example.com]
ok: [serverc.lab.example.com]
ok: [servera.lab.example.com]
ok: [serverd.lab.example.com]

TASK [Open the port for http] *****
changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]
changed: [servera.lab.example.com]
changed: [serverd.lab.example.com]

RUNNING HANDLER [restart http] *****
changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]
changed: [serverd.lab.example.com]
changed: [servera.lab.example.com]

PLAY RECAP *****
servera.lab.example.com : ok=9  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverb.lab.example.com : ok=9  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverc.lab.example.com : ok=9  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverd.lab.example.com : ok=9  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
sanjay devang 2025-04-25 > lab grade projects-review

Grading lab.

· Checking lab systems ..... SUCCESS
· Ensuring httpd service is started ..... PASS
· Ensuring firewall service is started ..... PASS
· Checking for Apache tune.conf ..... PASS
· Ensuring the web server is reachable on servera ..... PASS
· Ensuring the web server is reachable on serverb ..... PASS
· Ensuring the web server is reachable on serverc ..... PASS
· Ensuring the web server is reachable on serverd ..... PASS

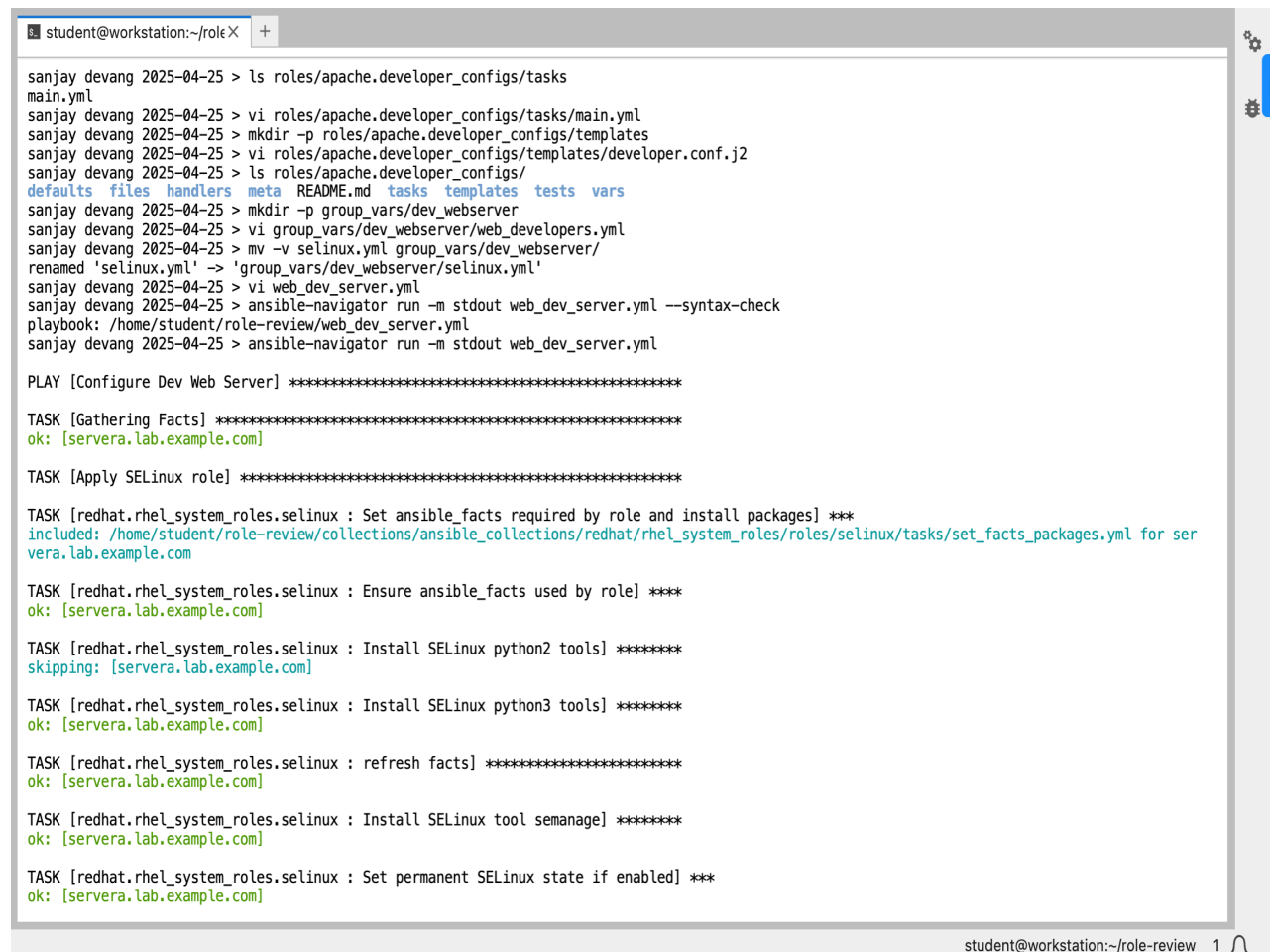
Overall lab grade: PASS

sanjay devang 2025-04-25 >

```

Figure 9 Web and firewall service configuration completed for all servers in the project review lab.

Note:The playbook enabled and started both the HTTPD and firewall services across servera, serverb, serverc, and serverd.lab.example.com. HTTP ports were opened, handlers triggered service restarts, and reachability for Apache was confirmed on all servers.



```

student@workstation:~/roleX +
sanjay devang 2025-04-25 > ls roles/apache.developer_configs/tasks
main.yml
sanjay devang 2025-04-25 > vi roles/apache.developer_configs/tasks/main.yml
sanjay devang 2025-04-25 > mkdir -p roles/apache.developer_configs/templates
sanjay devang 2025-04-25 > vi roles/apache.developer_configs/templates/developer.conf.j2
sanjay devang 2025-04-25 > ls roles/apache.developer_configs/
defaults files handlers meta README.md tasks templates tests vars
sanjay devang 2025-04-25 > mkdir -p group_vars/dev_webserver
sanjay devang 2025-04-25 > vi group_vars/dev_webserver/web_developers.yml
sanjay devang 2025-04-25 > mv -v selinux.yml group_vars/dev_webserver/
renamed 'selinux.yml' -> 'group_vars/dev_webserver/selinux.yml'
sanjay devang 2025-04-25 > vi web_dev_server.yml
sanjay devang 2025-04-25 > ansible-navigator run -m stdout web_dev_server.yml --syntax-check
playbook: /home/student/role-review/web_dev_server.yml
sanjay devang 2025-04-25 > ansible-navigator run -m stdout web_dev_server.yml

PLAY [Configure Dev Web Server] *****

TASK [Gathering Facts] *****
ok: [servera.lab.example.com]

TASK [Apply SELinux role] *****

TASK [redhat.rhel_system_roles.selinux : Set ansible_facts required by role and install packages] ***
included: /home/student/role-review/collections/ansible_collections/redhat/rhel_system_roles/roles/selinux/tasks/set_facts_packages.yml for ser
ver.a.lab.example.com

TASK [redhat.rhel_system_roles.selinux : Ensure ansible_facts used by role] ***
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.selinux : Install SELinux python2 tools] *****
skipping: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.selinux : Install SELinux python3 tools] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.selinux : refresh facts] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.selinux : Install SELinux tool semanage] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.selinux : Set permanent SELinux state if enabled] ***
ok: [servera.lab.example.com]

```

student@workstation:~/role-review 1

Figure 10 Starting web server configuration with SELinux adjustments using Ansible roles.

Note:The `web_dev_server.yml` playbook was executed to configure a development web server. The setup involved creating required directories, templates, and group variables, followed by applying the `redhat.rhel_system_roles.selinux` role to manage SELinux settings. Tasks like installing SELinux tools, refreshing facts, and setting SELinux policies were completed successfully for `servera.lab.example.com`.

```

student@workstation:~/role X +

TASK [apache.developer_configs : Create user accounts] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [apache.developer_configs : Give student access to all accounts] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [apache.developer_configs : Create content directory] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [apache.developer_configs : Create skeleton index.html if needed] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [apache.developer_configs : Set firewall port] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [apache.developer_configs : Copy Per-Developer Config files] *****
ok: [servera.lab.example.com] => (item={'username': 'jdoe', 'name': 'John Doe', 'user_port': 9081})
ok: [servera.lab.example.com] => (item={'username': 'jdoe2', 'name': 'Jane Doe', 'user_port': 9082})

TASK [Deploy production index.html] *****
changed: [servera.lab.example.com]

PLAY RECAP *****
servera.lab.example.com : ok=22  changed=1  unreachable=0  failed=0  skipped=16  rescued=0  ignored=0
sanjay devang 2025-04-25 > lab grade role-review

Grading lab.

  · Checking lab systems ..... SUCCESS
  · Ensuring httpd service is started ..... PASS
  · Ensuring firewalld service is started ..... PASS
  · Verify web content for servera ..... PASS
  · Verify web content for user jdoe on servera ..... PASS
  · Verify web content for user jdoe2 on servera ..... PASS

Overall lab grade: PASS

sanjay devang 2025-04-25 >

```

Figure 11 Developer configuration tasks completed for user accounts and web content in role review lab.

Note:The `role-review` lab executed multiple tasks including creating developer user accounts, setting up directories, firewall ports, and content files for users `jdoe1` and `jdoe2`. SELinux adjustments and production index deployment were also successfully performed.


```
student@workstation:~/troubleshoot-review$ ansible webservers -u devops -b -m command -a "systemctl status httpd"
PLAY RECAP *****
serverb.lab.example.com : ok=8   changed=0   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0
sanjay devang 2025-04-25 > ansible webservers -u devops -b -m command -a "systemctl status httpd"
serverb.lab.example.com | CHANGED | rc=0 >>
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Fri 2025-04-25 15:41:19 EDT; 32min ago
     Docs: man:httpd.service(8)
  Main PID: 769 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 213 (limit: 5588)
   Memory: 34.3M
      CPU: 1.435s
   CGroup: /system.slice/httpd.service
           └─769 /usr/sbin/httpd -DFOREGROUND
             └─796 /usr/sbin/httpd -DFOREGROUND
               └─797 /usr/sbin/httpd -DFOREGROUND
                 └─798 /usr/sbin/httpd -DFOREGROUND
                   └─799 /usr/sbin/httpd -DFOREGROUND

Apr 25 15:41:18 serverc.lab.example.com systemd[1]: Starting The Apache HTTP Server...
Apr 25 15:41:19 serverc.lab.example.com httpd[769]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using serverc.lab.example.com. Set the 'ServerName' directive globally to suppress this message
Apr 25 15:41:19 serverc.lab.example.com systemd[1]: Started The Apache HTTP Server.
Apr 25 15:41:19 serverc.lab.example.com httpd[769]: Server configured, listening on: port 80
sanjay devang 2025-04-25 > ansible webservers -u devops -b -m command -a "firewall-cmd --list-all"
serverb.lab.example.com | CHANGED | rc=0 >>
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: eth0
  sources:
  services: cockpit dhcpv6-client http https ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
sanjay devang 2025-04-25 > lab grade troubleshoot-review
```

Figure 12 HTTPD service status verified and firewall settings inspected during troubleshooting review.

Note: In the troubleshoot-review lab, system commands confirmed that the Apache HTTP server (httpd) was active and running on `serverb.lab.example.com`. Firewall rules were listed to verify open services like HTTP, HTTPS, and SSH.

```

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]

TASK [Install web server packages] *****
changed: [serverb.lab.example.com] => (item=httpd)
changed: [serverb.lab.example.com] => (item=mod_ssl)

TASK [Install httpd config files] *****
changed: [serverb.lab.example.com]

TASK [Create ssl certificate] *****
changed: [serverb.lab.example.com]

TASK [Start and enable web service] *****
changed: [serverb.lab.example.com]

TASK [Open firewall ports for HTTP and HTTPS] *****
changed: [serverb.lab.example.com] => (item=http)
changed: [serverb.lab.example.com] => (item=https)

TASK [Create directory for secure web content] *****
changed: [serverb.lab.example.com]

TASK [Deliver content] *****
changed: [serverb.lab.example.com]

RUNNING HANDLER [Restart services] *****
changed: [serverb.lab.example.com]

PLAY RECAP *****
serverb.lab.example.com : ok=9  changed=8  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
sanjay devang 2025-04-25 > lab grade troubleshoot-review

Grading lab.
  • Checking lab systems ..... SUCCESS
  • Ensuring httpd service is started ..... PASS
  • Ensuring firewalld service is started ..... PASS
  • Ensuring the web server is reachable on serverb ..... PASS

Overall lab grade: PASS

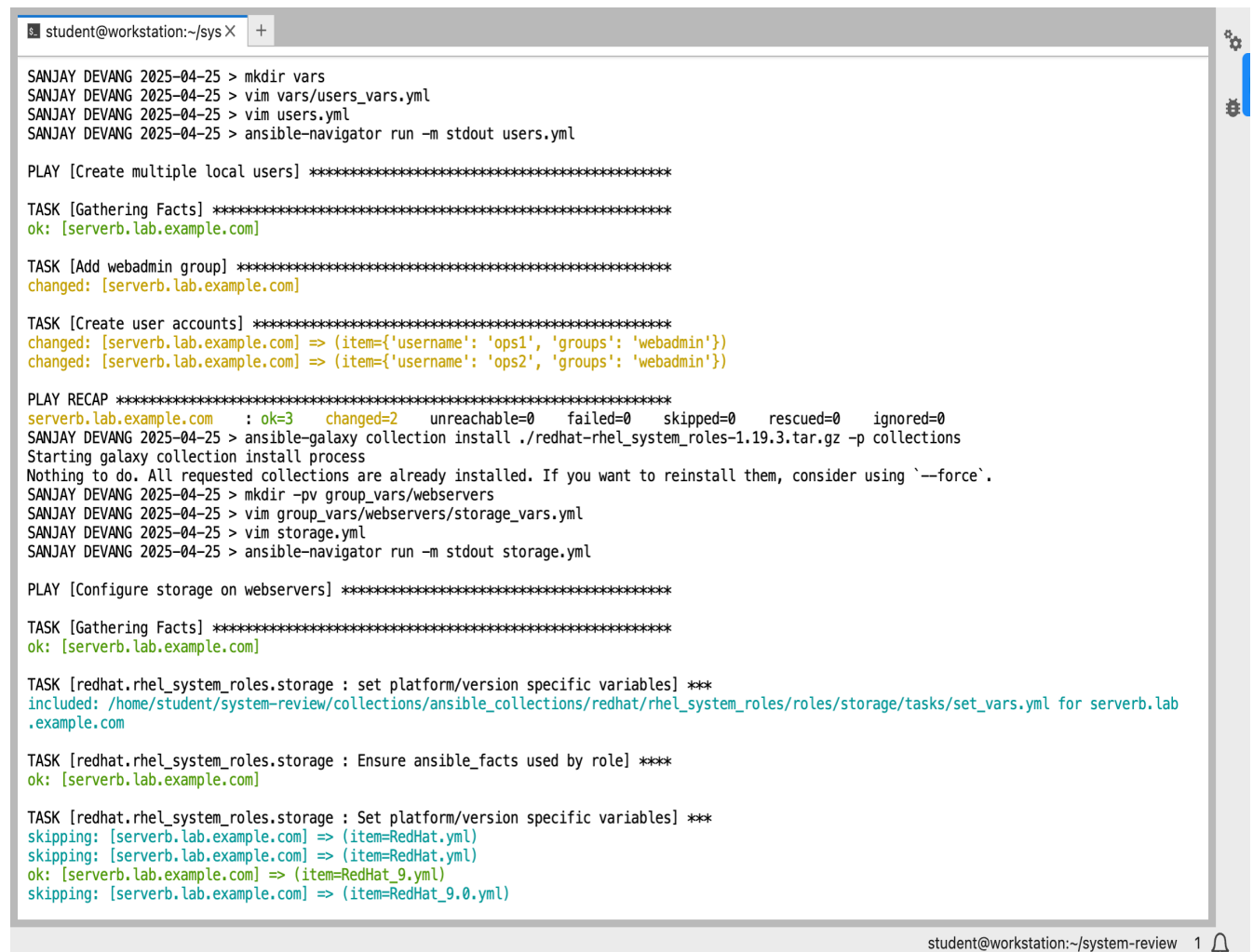
sanjay devang 2025-04-25 > 

```

student@workstation:~/troubleshoot-review 1

Figure 13 Successful web server setup and firewall configuration in troubleshoot review lab.

Note:The troubleshoot-review lab tasks included installing HTTPD and SSL packages, creating SSL certificates, opening firewall ports, and deploying web content securely. Services were restarted as needed, and all validations for web server functionality and firewall accessibility passed.



```

student@workstation:~/sys X +

SANJAY DEVANG 2025-04-25 > mkdir vars
SANJAY DEVANG 2025-04-25 > vim vars/users_vars.yml
SANJAY DEVANG 2025-04-25 > vim users.yml
SANJAY DEVANG 2025-04-25 > ansible-navigator run -m stdout users.yml

PLAY [Create multiple local users] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]

TASK [Add webadmin group] *****
changed: [serverb.lab.example.com]

TASK [Create user accounts] *****
changed: [serverb.lab.example.com] => (item={'username': 'ops1', 'groups': 'webadmin'})
changed: [serverb.lab.example.com] => (item={'username': 'ops2', 'groups': 'webadmin'})

PLAY RECAP *****
serverb.lab.example.com : ok=3  changed=2  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SANJAY DEVANG 2025-04-25 > ansible-galaxy collection install ./redhat-rhel_system_roles-1.19.3.tar.gz -p collections
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall them, consider using `--force`.
SANJAY DEVANG 2025-04-25 > mkdir -pv group_vars/webserver
SANJAY DEVANG 2025-04-25 > vim group_vars/webserver/storage_vars.yml
SANJAY DEVANG 2025-04-25 > vim storage.yml
SANJAY DEVANG 2025-04-25 > ansible-navigator run -m stdout storage.yml

PLAY [Configure storage on webserver] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.storage : set platform/version specific variables] ***
included: /home/student/system-review/collections/ansible_collections/redhat/rhel_system_roles/roles/storage/tasks/set_vars.yml for serverb.lab.example.com

TASK [redhat.rhel_system_roles.storage : Ensure ansible_facts used by role] ***
ok: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.storage : Set platform/version specific variables] ***
skipping: [serverb.lab.example.com] => (item=RedHat.yml)
skipping: [serverb.lab.example.com] => (item=RedHat.yml)
ok: [serverb.lab.example.com] => (item=RedHat_9.yml)
skipping: [serverb.lab.example.com] => (item=RedHat_9.0.yml)

```

student@workstation:~/system-review 1

Figure 14 Local user creation and system roles setup for web servers in system review lab.

Note:The `users.yml` playbook created local users and added them to the `webadmin` group on `serverb.lab.example.com`. System roles were configured using the `rhel_system_roles.storage` role to ensure platform-specific settings. Tasks completed successfully, with Ansible Galaxy collections used to manage role dependencies.

```

student@workstation:~/sys X +
skipping: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Ensure initscripts network file dependency is present] ***
skipping: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Configure networking connection profiles] ***
changed: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Configure networking state] *****
skipping: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Show stderr messages for the network_connections] ***
ok: [serverb.lab.example.com] => {
  "_network_connections_result.stderr_lines": [
    "[002] <info> #0, state:None persistent_state:present, 'eth1': add connection eth1, 52f89866-f0a7-4051-b24c-02544a8611dd"
  ]
}

TASK [redhat.rhel_system_roles.network : Show debug messages for the network_connections] ***
skipping: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Show debug messages for the network_state] ***
skipping: [serverb.lab.example.com]

TASK [redhat.rhel_system_roles.network : Re-test connectivity] *****
ok: [serverb.lab.example.com]

PLAY RECAP *****
serverb.lab.example.com : ok=10 changed=1 unreachable=0 failed=0 skipped=12 rescued=0 ignored=0
SANJAY DEVANG 2025-04-25 > lab grade system-review

Grading lab.

· Checking lab systems ..... SUCCESS
· Verify rhelver package is installed ..... PASS
· Verify ops1 and ops2 users are available ..... PASS
· Verify content-lv LV is available and mounted ..... PASS
· Verify logs-lv LV is available and mounted ..... PASS
· Verify the cron job is configured ..... PASS
· Verify the second NIC is configured ..... PASS

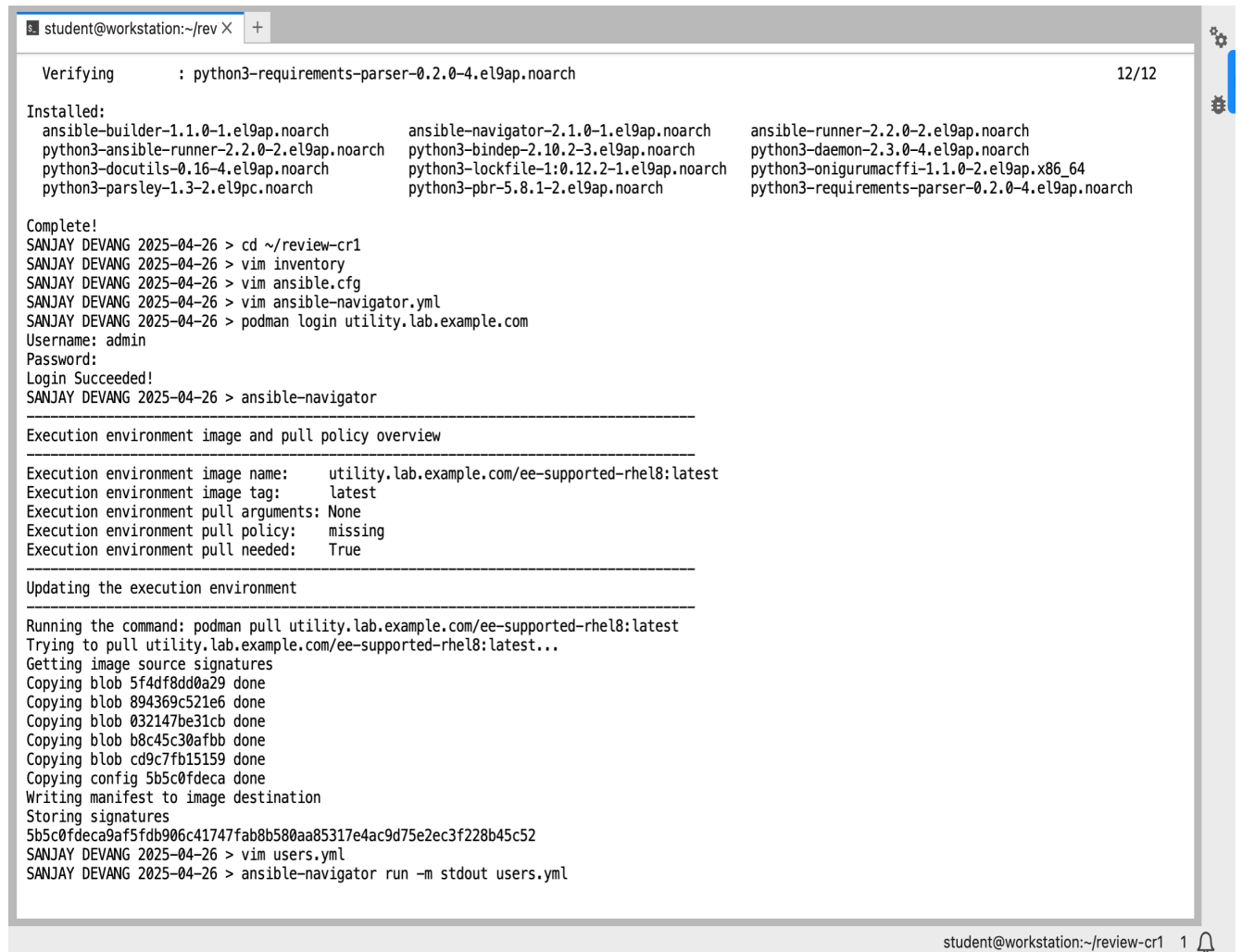
Overall lab grade: PASS

SANJAY DEVANG 2025-04-25 >

```

Figure 15 Network configuration and system validation completed for system review lab.

Note: Using the `rhel_system_roles.network` role, network settings were applied and verified on `serverb.lab.example.com`. Tasks included configuring `eth1`, checking connection states, and validating system storage (LVM and logs). All system checks passed, confirming correct setup of networking and storage components



```

student@workstation:~/rev X +
Verifying      : python3-requirements-parser-0.2.0-4.el9ap.noarch 12/12

Installed:
ansible-builder-1.1.0-1.el9ap.noarch      ansible-navigator-2.1.0-1.el9ap.noarch      ansible-runner-2.2.0-2.el9ap.noarch
python3-ansible-runner-2.2.0-2.el9ap.noarch python3-bindep-2.10.2-3.el9ap.noarch      python3-daemon-2.3.0-4.el9ap.noarch
python3-docutils-0.16-4.el9ap.noarch      python3-lockfile-1:0.12.2-1.el9ap.noarch  python3-onigurumacffi-1.1.0-2.el9ap.x86_64
python3-parsley-1.3-2.el9pc.noarch        python3-pbr-5.8.1-2.el9ap.noarch          python3-requirements-parser-0.2.0-4.el9ap.noarch

Complete!
SANJAY DEVANG 2025-04-26 > cd ~/review-cr1
SANJAY DEVANG 2025-04-26 > vim inventory
SANJAY DEVANG 2025-04-26 > vim ansible.cfg
SANJAY DEVANG 2025-04-26 > vim ansible-navigator.yml
SANJAY DEVANG 2025-04-26 > podman login utility.lab.example.com
Username: admin
Password:
Login Succeeded!
SANJAY DEVANG 2025-04-26 > ansible-navigator

-----
Execution environment image and pull policy overview
-----
Execution environment image name:  utility.lab.example.com/ee-supported-rhel8:latest
Execution environment image tag:   latest
Execution environment pull arguments: None
Execution environment pull policy:  missing
Execution environment pull needed:  True
-----

Updating the execution environment
-----
Running the command: podman pull utility.lab.example.com/ee-supported-rhel8:latest
Trying to pull utility.lab.example.com/ee-supported-rhel8:latest...
Getting image source signatures
Copying blob 5f4df8dd0a29 done
Copying blob 894369c521e6 done
Copying blob 032147be31cb done
Copying blob b8c45c30afbb done
Copying blob cd9c7fb15159 done
Copying config 5b5c0fdeca done
Writing manifest to image destination
Storing signatures
5b5c0fdeca9af5fdb906c41747fab8b580aa85317e4ac9d75e2ec3f228b45c52
SANJAY DEVANG 2025-04-26 > vim users.yml
SANJAY DEVANG 2025-04-26 > ansible-navigator run -m stdout users.yml

```

Figure 16 Setting up execution environment and pulling container image for review lab.

Note: The system installed required Ansible and Python packages and logged into the utility registry. An execution environment image `utility.lab.example.com/ee-supported-rhel8:latest` was pulled using `podman` to support isolated Ansible execution. Environment setup completed successfully, preparing for running the `users.yml` playbook.

```

TASK [Verify the sam user exists] *****
ok: [servera.lab.example.com]
ok: [serverb.lab.example.com]

TASK [Sam was created] *****
ok: [servera.lab.example.com] => {
  "msg": "Sam was created"
}
ok: [serverb.lab.example.com] => {
  "msg": "Sam was created"
}

TASK [Output sam user status to file] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

PLAY RECAP *****
servera.lab.example.com : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
serverb.lab.example.com : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
SANJAY DEVANG 2025-04-26 > lab grade review-cr1

Grading lab.

• Checking lab systems ..... SUCCESS
• Ensuring ansible-navigator package is installed on workstation ..... PASS
• Ensuring inventory file is present ..... PASS
• Ensuring ansible.cfg file is present ..... PASS
• Ensuring ansible-navigator.yml file is present ..... PASS
• Ensuring the ee-supported-rhel8 image is present ..... PASS
• Ensuring users.yml playbook is present ..... PASS
• Verify joe and sam users are available on servera ..... PASS
• Verify joe and sam users are available on serverb ..... PASS
• Ensuring httpd package is installed on servera ..... PASS
• Ensuring httpd package is installed on serverb ..... PASS
• Ensuring mariadb-server package is installed on servera ..... PASS
• Ensuring mariadb-server package is installed on serverb ..... PASS
• Ensuring redis package is installed on serverb ..... PASS
• Ensuring successful verify_user.yml run on servera ..... PASS
• Ensuring successful verify_user.yml run on serverb ..... PASS

Overall lab grade: PASS

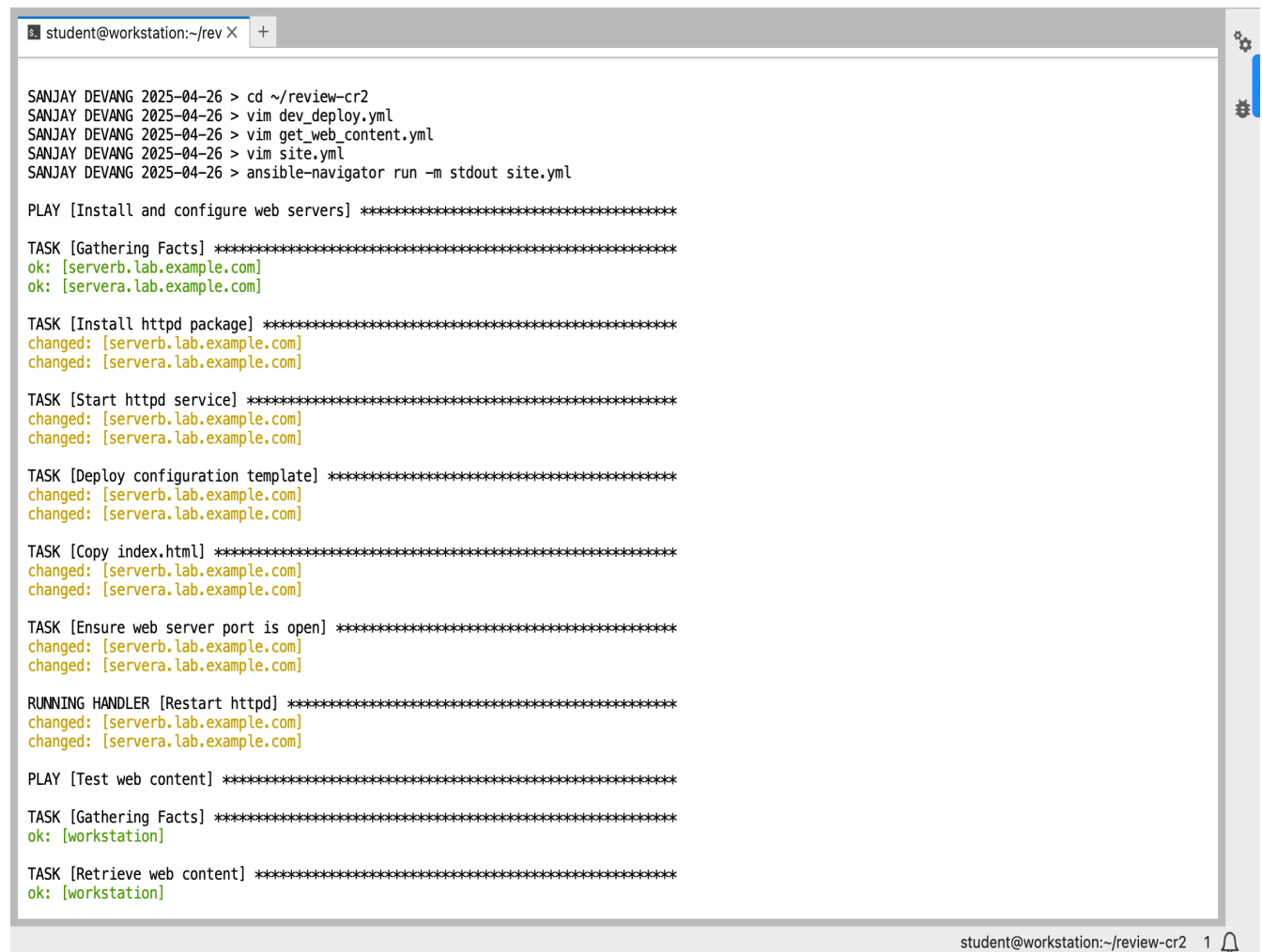
SANJAY DEVANG 2025-04-26 > █

```

student@workstation:~/review-cr1 1

Figure 17 User verification tasks completed and graded successfully in review-cr1 lab.

Note:The `users.yml` playbook verified the creation of the `sam` user on `servera` and `serverb.lab.example.com`, outputting confirmation messages. The `review-cr1` lab grading confirmed all requirements were met, including package installation, environment setup, and user creation.



```
student@workstation:~/rev X +

SANJAY DEVANG 2025-04-26 > cd ~/review-cr2
SANJAY DEVANG 2025-04-26 > vim dev_deploy.yml
SANJAY DEVANG 2025-04-26 > vim get_web_content.yml
SANJAY DEVANG 2025-04-26 > vim site.yml
SANJAY DEVANG 2025-04-26 > ansible-navigator run -m stdout site.yml

PLAY [Install and configure web servers] *****

TASK [Gathering Facts] *****
ok: [serverb.lab.example.com]
ok: [servera.lab.example.com]

TASK [Install httpd package] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

TASK [Start httpd service] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

TASK [Deploy configuration template] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

TASK [Copy index.html] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

TASK [Ensure web server port is open] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

RUNNING HANDLER [Restart httpd] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

PLAY [Test web content] *****

TASK [Gathering Facts] *****
ok: [workstation]

TASK [Retrieve web content] *****
ok: [workstation]
```

student@workstation:~/review-cr2 1

Figure 18 Web server installation, configuration, and content verification during review-cr2 lab.

Note:In the review-cr2 lab, the `site.yml` playbook installed HTTPD, deployed a configuration template, copied the `index.html`, and ensured the web port was open on `serverb.lab.example.com`. The service was restarted, and web content retrieval from the workstation was verified successfully, indicating the server was properly configured.


```

student@workstation:~/rev X +
TASK [Copy index.html] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

TASK [Ensure web server port is open] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

RUNNING HANDLER [Restart httpd] *****
changed: [serverb.lab.example.com]
changed: [servera.lab.example.com]

PLAY [Test web content] *****

TASK [Gathering Facts] *****
ok: [workstation]

TASK [Retrieve web content] *****
ok: [workstation]

PLAY RECAP *****
servera.lab.example.com : ok=7  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverb.lab.example.com : ok=7  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
workstation             : ok=2  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SANJAY DEVANG 2025-04-26 > lab grade review-cr2

Grading lab.

  • Checking lab systems ..... SUCCESS
  • Ensuring httpd package is installed on servera ..... PASS
  • Ensuring httpd package is installed on serverb ..... PASS
  • Ensuring dev_deploy.yml playbook file is present ..... PASS
  • Ensuring get_web_content.yml playbook file is present ..... PASS
  • Ensuring site.yml playbook file is present ..... PASS
  • Ensuring index.html file is present on servera ..... PASS
  • Ensuring index.html file is present on serverb ..... PASS
  • Ensuring vhost.conf file is present on servera ..... PASS
  • Ensuring vhost.conf file is present on serverb ..... PASS
  • Ensuring the web server is reachable on servera ..... PASS
  • Ensuring the web server is reachable on serverb ..... PASS

Overall lab grade: PASS
SANJAY DEVANG 2025-04-26 >

```

Figure 19 Review-cr2 lab grading shows successful web server deployment and validation.

Note:The final grading for review-cr2 lab confirmed that all tasks were completed: HTTPD package installation, playbook file validation, index and configuration file deployment, and web server accessibility. The web server was reachable from the workstation.

The screenshot shows a JupyterLab interface with a terminal window. The terminal displays the output of an Ansible playbook run. The first part shows the execution of the `log-rotate.yml` playbook, which successfully schedules a recurring cron job for log rotation on `servera.lab.example.com`. The second part shows the execution of the `site.yml` playbook, which begins configuring storage using the `redhat.rhel_system_roles.storage` role. The terminal output includes status messages for various tasks, such as gathering facts, checking if the crontab file exists, and setting platform-specific variables.

```

student@workstation:~/rev X
SANJAY DEVANG 2025-04-26 > ansible-navigator run -m stdout log-rotate.yml

PLAY [Recurring cron job] *****

TASK [Gathering Facts] *****
ok: [servera.lab.example.com]

TASK [Crontab file exists] *****
changed: [servera.lab.example.com]

PLAY RECAP *****
servera.lab.example.com : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
SANJAY DEVANG 2025-04-26 > ssh devops@servera "cat /etc/cron.d/rotate_web"
#Ansible: Rotate HTTPD logs
0 0 * * * devops logrotate -f /etc/logrotate.d/httpd
SANJAY DEVANG 2025-04-26 > vim site.yml
SANJAY DEVANG 2025-04-26 > ansible-navigator run -m stdout --pae false site.yml --vault-id @prompt
Vault password (default):

PLAY [Configure storage on webserver] *****

TASK [Gathering Facts] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.storage : set platform/version specific variables] *****
included: /home/student/review-cr3/collections/ansible_collections/redhat/rhel_system_roles/roles/storage/tasks/set_vars.yml for servera.lab.example.com

TASK [redhat.rhel_system_roles.storage : Ensure ansible_facts used by role] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.storage : Set platform/version specific variables] *****
skipping: [servera.lab.example.com] => (item=RedHat.yml)
skipping: [servera.lab.example.com] => (item=RedHat.yml)
ok: [servera.lab.example.com] => (item=RedHat_9.yml)
skipping: [servera.lab.example.com] => (item=RedHat_9.0.yml)

TASK [redhat.rhel_system_roles.storage : define an empty list of pools to be used in testing] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.storage : define an empty list of volumes to be used in testing] *****
ok: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.storage : include the appropriate provider tasks] *****

```

Figure 20 Cron job setup and initial storage role configuration during review-cr3 lab.

Note: The `log-rotate.yml` playbook successfully scheduled a recurring cron job for log rotation on `servera.lab.example.com`. Following that, the `site.yml` playbook began configuring storage using the `redhat.rhel_system_roles.storage` role, ensuring facts were gathered and platform-specific variables were set. The environment is prepared for further storage setup tasks.

```

student@workstation:~/rev X +
skipping: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.network : Show debug messages for the network_state] *****
skipping: [servera.lab.example.com]

TASK [redhat.rhel_system_roles.network : Re-test connectivity] *****
ok: [servera.lab.example.com]

PLAY [Recurring cron job] *****

TASK [Gathering Facts] *****
ok: [servera.lab.example.com]

TASK [Crontab file exists] *****
ok: [servera.lab.example.com]

PLAY RECAP *****
servera.lab.example.com : ok=37  changed=0  unreachable=0  failed=0  skipped=24  rescued=0  ignored=0

SANJAY DEVANG 2025-04-26 > lab grade review-cr3

Grading lab.

  · Checking lab systems ..... SUCCESS
  · Ensuring vg_web volume group exists ..... PASS
  · Verifying vg_web volume group physical volume ..... PASS
  · Verifying that lv_content logical volume exists ..... PASS
  · Verifying lv_content logical volume size ..... PASS
  · Verifying lv_content logical volume mount ..... PASS
  · Verifying lv_content logical volume file system ..... PASS
  · Verifying that lv_uploads logical volume exists ..... PASS
  · Verifying lv_uploads logical volume size ..... PASS
  · Verifying lv_uploads logical volume mount ..... PASS
  · Verifying lv_uploads logical volume file system ..... PASS
  · Verify developer user exists ..... PASS
  · Verify developer user sudoers rule ..... PASS
  · Verify developer user password ..... PASS
  · Verify eth1 interface IP address ..... PASS
  · Ensuring rotate_web cronjob file is present ..... PASS
  · Ensuring site.yml playbook is present ..... PASS

Overall lab grade: PASS

SANJAY DEVANG 2025-04-26 >

```

Figure 21 Final grading results for review-cr3 lab confirming storage and system setup.

Note:The review-cr3 lab successfully validated the creation and mounting of multiple logical volumes, volume groups, cron job scheduling, and developer user configurations. The playbook ensured all critical system components were correctly provisioned.

```

student@workstation:~/rev X +
UpdateHostkeys is disabled because the host key is not trusted.
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Sat Apr 26 20:30:23 2025
[student@workstation ~]$ lab start review-cr4

Starting lab.

  · Checking lab systems ..... SUCCESS
  · Adding exercise content ..... SUCCESS
  · Ensuring required packages are installed ..... SUCCESS

[student@workstation ~]$ PS1='SANJAY DEVANG \D{%Y-%m-%d} > '
SANJAY DEVANG 2025-04-26 > cd ~/review-cr4
SANJAY DEVANG 2025-04-26 > mkdir -v roles
mkdir: created directory 'roles'
SANJAY DEVANG 2025-04-26 > cd roles
SANJAY DEVANG 2025-04-26 > ansible-galaxy role init ansible-httpd
- Role ansible-httpd was created successfully
SANJAY DEVANG 2025-04-26 > cd ..
SANJAY DEVANG 2025-04-26 > vim roles/ansible-httpd/defaults/main.yml
SANJAY DEVANG 2025-04-26 > cp -v templates/httpd.conf.j2 roles/ansible-httpd/templates/
'templates/httpd.conf.j2' -> 'roles/ansible-httpd/templates/httpd.conf.j2'
SANJAY DEVANG 2025-04-26 > vim roles/ansible-httpd/tasks/main.yml
SANJAY DEVANG 2025-04-26 > cp -v files/index.html roles/ansible-httpd/files/
'files/index.html' -> 'roles/ansible-httpd/files/index.html'
SANJAY DEVANG 2025-04-26 > vim roles/ansible-httpd/handlers/main.yml
SANJAY DEVANG 2025-04-26 > vim roles/ansible-httpd/meta/main.yml
SANJAY DEVANG 2025-04-26 > vim roles/ansible-httpd/README.md
SANJAY DEVANG 2025-04-26 > rm -rfv roles/ansible-httpd/vars/
removed 'roles/ansible-httpd/vars/main.yml'
removed directory 'roles/ansible-httpd/vars/'
SANJAY DEVANG 2025-04-26 > rm -rfv roles/ansible-httpd/tests/
removed 'roles/ansible-httpd/tests/inventory'
removed 'roles/ansible-httpd/tests/test.yml'
removed directory 'roles/ansible-httpd/tests/'
SANJAY DEVANG 2025-04-26 > vim site.yml
SANJAY DEVANG 2025-04-26 > ansible-navigator run -m stdout site.yml

PLAY [Apply the ansible-httpd role] *****

TASK [Gathering Facts] *****

```

student@workstation:~/review-cr4 1

Figure 22 Initialization of custom Ansible role `ansible-httpd` for `review-cr4` lab.

Note: The `review-cr4` lab involved creating a new Ansible role named `ansible-httpd` using `ansible-galaxy`. Templates, tasks, files, and metadata were organized appropriately, and unnecessary directories like `tests` were removed. The `site.yml` playbook was prepared to apply the role and automate the web server setup.

```

student@workstation:~/rev X +

changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]

TASK [ansible-httpd : Deploy index.html file] *****
changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]

TASK [ansible-httpd : Web port is open] *****
changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]

RUNNING HANDLER [ansible-httpd : Restart httpd] *****
changed: [serverb.lab.example.com]
changed: [serverc.lab.example.com]

PLAY RECAP *****
serverb.lab.example.com : ok=7  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
serverc.lab.example.com : ok=7  changed=6  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
SANJAY DEVANG 2025-04-26 > lab grade review-cr4

Grading lab.

  • Checking lab systems ..... SUCCESS
  • Verify the /home/student/review-cr4/roles/ansible-httpd directory exists ..... PASS
  • Verify the defaults/main.yml file exists ..... PASS
  • Verify the templates/httpd.conf.j2 file exists ..... PASS
  • Verify the handlers/main.yml file exists ..... PASS
  • Verify the meta/main.yml file exists ..... PASS
  • Verify empty vars directory inside the role was removed ..... PASS
  • Verify empty tests directory inside the role was removed ..... PASS
  • Verify the 'site.yml' playbook exists ..... PASS
  • Verify the templates/httpd.conf.j2 file exists ..... PASS
  • Verify the httpd service is running on serverb ..... PASS
  • Verify the httpd service is running on serverc ..... PASS
  • Verify the /etc/httpd/conf/httpd.conf file was deployed on serverb ..... PASS
  • Verify the /etc/httpd/conf/httpd.conf file was deployed on serverc ..... PASS
  • Verify the /var/www/html/index.html file was deployed on serverb ..... PASS
  • Verify the /var/www/html/index.html file was deployed on serverc ..... PASS
  • Verify the http service is reachable on serverb ..... PASS
  • Verify the http service is reachable on serverc ..... PASS

Overall lab grade: PASS

SANJAY DEVANG 2025-04-26 >

```

Figure 23 Successful deployment and validation of the custom ansible-httpd role in review-cr4 lab.

Note:The `ansible-httpd` role correctly deployed configuration and content files, ensured HTTPD service functionality, and opened the web server port across multiple servers.

Summary of Lessons Learned

In the course of participating in this set of labs, I have substantially improved my abilities in the areas of Linux system administration, service setup, and automation through the utilisation of Ansible. For the purpose of installing packages, configuring and managing services such as HTTPD, MariaDB, and firewalld, and automating system operations such as setting up recurring cron jobs, I learnt how to efficiently create and execute Ansible playbooks.

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Through my work with Red Hat System Roles, I was able to get an understanding of how to automate more complex settings for SELinux policies, storage management, and networking. As part of my training, I collected information about the system, applied settings that were specific to the platform, and validated network connectivity using automation. With the help of Ansible Vault, I was also able to teach myself how to safely manage sensitive information while playbooks were being executed.

I started from scratch and constructed unique Ansible roles, ensuring that tasks, handlers, templates, files, and variables were appropriately organised. I then applied these roles to configure web servers across many nodes across the network. Through this experience, I gained knowledge about role-based automation development as well as the significance of modular and reusable code in real-world settings.

My ability to troubleshoot was further improved by the fact that I was able to diagnose service misconfigurations, verify system statuses, validate SSL certificates, and make certain that services were accessible across the network. My awareness of contemporary DevOps approaches was further enhanced by the fact that I worked with tools such as podman to handle containerised execution environments.

It was determined through grading validations that each phase of the deployment process, such as user management, logical volume management, firewall rules, and web server availability, had been successfully completed. As a result of participating in these laboratories, I was able to acquire practical experience in the configuration of enterprise-grade systems, the creation of efficient Ansible automation, the management of secure online deployments, and the verification that systems are audit-ready and compliant.